Journal of Physics:
Conference Series

PAPER • OPEN ACCESS

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To cite this article: Alexander Neubauer *et al* 2025 *J. Phys.: Conf. Ser.* **3140** 052023

CISBAT 2025 IOP Publishing
Journal of Physics: Conference Series **3140** (2025) 052023 doi:10.1088/1742-6596/3140/5/052023

Influence of occupancy data on heating and electricity load forecasting using machine learning

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Abstract. Hotels are highly energy-intensive buildings due to continuous operation and fluctuating occupancy. Accurate forecasting of heating and electricity demand is essential for optimising combined heat and power (CHP) systems and enabling grid responsive control strategies. While previous research has investigated the influence of occupancy information in energy forecasting, its role in LSTM-based models for different hotel usage types remains unexplored. This paper investigates the influence of occupancy data on heating and electricity load forecasting with different usage types using real data from two German hotels. Forecasts are made for heating and electricity loads in various zones, such as the swimming pool and the guest room wing. Long short-term memory (LSTM) and encoder-decoder (ED) models are applied. The results show that the inclusion of occupancy data slightly reduces the average normalized root mean squared error (NRMSE) for heating forecasts from 4.91 % to 4.90 %, and for electricity forecasts from 7.20 % to 7.07 %. These averages are based on the LSTM-based encoder-decoder (ED) model, which consistently achieved lower forecast errors than LSTM. Both configurations with and without time features were considered. The permutation feature importance (PFI) shows that occupancy contributes up to 14.7 % in heating forecasts. A relationship is observed whereby higher feature importance for occupancy often coincides with stronger improvements or degradations in forecast accuracy, depending on the forecast objective.

1. Introduction

Building operations are responsible for 26% of world energy-related emissions and 30% of global total energy consumption[1]. Hotels are among the most energy-intensive buildings, mainly due to their 24-hour operation, multifunctionality, and high visitor occupancy rates [2]. Heating and electricity forecasts are particularly useful for hotels, where combined heat and power (CHP) units are often installed. Accurate forecasts help optimize CHP unit control, increasing economic efficiency by maximizing on-site electricity use.

Existing research has used features such as weather and time data to forecast the loads of hotels, often using machine learning techniques. For example, Ahmad et al. [3] reported a comparison of Artificial Neural Networks (ANN) and Random Forest (RF) models to forecast Heating, Ventilation and Air Conditioning (HVAC) electricity consumption not only with environmental and time features, but also with occupancy information. They found that while the inclusion of this occupancy information marginally improved prediction accuracy, the gains were relatively modest for both ANN and RF techniques, proving their suitability for building energy management systems. Borowski and Zwolińska [4] compared Support Vector Machine

(SVM) and ANN models for cooling load forecasting in a hotel in southern Poland and found that occupancy had a moderate impact on prediction accuracy. Similarly, Casteleiro-Roca et al. [5] utilized a hybrid model Multi-Layer Perceptron (MLP), K-Means clustering, and Least Squares Support Vector Regression to forecast short-term energy demand in a Tenerife Canary Island luxury hotel. However, in all of the above mentioned works, they have different limitations: [3] primarily focus on individual systems (heating or cooling), [5] do not account for variable occupancy, while Borowski and Zwolińska [4] did not effectively capture occupancy's non-linear effects. LSTM networks have demonstrated superior performance over traditional models like MLPs or SVRs [6] in time-series energy forecasting, due to their ability to model temporal dependencies. In this paper, we study the effect of occupancy on heating and electricity load forecasts using actual datasets from two German hotels with different usage patterns. The datasets include consumption data with hourly resolution, including heating and electricity load, occupancy, weather data, and time characteristics. It is analysed whether the Permutation Feature Importance (PFI) is a good indicator to determine the influence of occupancy on load forecasts. It is also investigated whether the impact of occupancy is different for various types of usage. We employ state-of-the-art machine learning models, specifically LSTM models and sequence-to-sequence (seq2seq) encoder-decoder (ED) models, that are designed to capture temporal dependencies in high-resolution time-series data. This study assumes that including occupancy data will have a quantifiable effect on load forecasting depending on usage type.

2. Methods
First, the two ML models used, LSTM and the seq2seq ED model, are described in Section 2.1. In Section 2.2, the datasets are explained.

2.1. ML Models
Both machine learning models in this study operate on hourly time series data and use the past 24 hours of input features to predict the hour future load values of the next 24 hours. The past features include weather variables, occupancy and time features such as hour of the day, day of the week and week of the year. While the LSTM model generates 24 step-ahead forecasts based solely on past inputs, the ED model also incorporates future features such as weather forecasts and expected occupancy for the corresponding forecast horizon. This allows the ED architecture to better model future patterns by using information that is available or predictable at the time of forecasting.

Long-Short-Term Memory (LSTM)
LSTM networks are a specialised form of recurrent neural networks (RNNs) designed to overcome the limitations of standard RNNs in capturing long-term dependencies. By incorporating memory cells with input, output and forgetting gates, LSTMs can effectively store and update relevant information over long sequences. In energy forecasting, they are often used to learn complex temporal patterns from historical energy load data, weather data, time data and other external factors, enabling more accurate forecasts over different time horizons.

Encoder-Decoder (ED)
ED models use an encoder-decoder architecture that maps an input sequence to an output sequence. The encoder processes historical energy data into a summarised representation, while the decoder uses this representation to make predictions for future energy consumption. In this architecture, the encoder processes historical energy data, captures complex temporal patterns and compresses the information into a fixed-length context vector. This vector is then passed to the decoder, which creates a sequence of future energy forecasts. The Seq2Seq model in this work

is a framework that uses two LSTM networks for both the encoder and the decoder to convert an input sequence into an output sequence. This LSTM-based Seq2Seq approach is particularly effective in energy forecasting because it can handle variable-length input sequences and produce multi-stage forecasts, while its gating mechanisms help to preserve long-term dependencies in the time series.

2.2. Dataset

Heating and electricity load were collected from energy metres in two hotels in Germany. The measurement period spans from February 1, 2022, to September 30, 2024, with a temporal resolution of 1 h. The weather data was obtained from Open Meteo [7]. While the daily occupancy data were provided by the hotels' internal management systems. To analyse heating loads $\dot{Q}$, multiple usage types with distinct consumption profiles were considered: the entire hotel (Hotel) consumption, the guest room wing (GRW) and the swimming pool (SP). In contrast, the electricity load P_{el} was evaluated only at the whole building level. This differentiation enables an assessment of whether occupancy (Occu) influences forecasting performance differently across specific usage areas.

Table 1 shows the statistical properties (min, max, standard deviation (std) and median) of the features and labels of the load profiles for both hotels, based on the complete dataset collected between February 1, 2022, and September 30, 2024. The features include occupancy, weather variables—namely outdoor temperature T_{out} and Global Horizontal Irradiance (GHI)—as well as time features such as hour of the day, day of the week, and week of the year.

Table 1. Statistical properties of the various datasets for the features and labels

Hotel	Value	$T_{\text{out}}/°\text{C}$	$\text{GHI}/\text{W}\,\text{m}^{-2}$	Occu/1	$\dot{Q}_{\text{SP}}/\text{kW}$	$\dot{Q}_{\text{GRW}}/\text{kW}$	$\dot{Q}_{\text{Hotel}}/\text{kW}$	P_{el}/kW
1	Min	-10.9	0	40	0	0	77.9	214
	Max	36.9	877	916	1079	464.6	1745.8	1314
	Std	7.7	207	238	198.2	109.8	300.4	190.9
	Median	11.1	13	495	361.3	127.9	737.9	720.4
2	Min	-13.4	0	17	0	0	0	225.7
	Max	31.7	898	475	380.4	3561.2	5061.1	1216
	Std	7.9	213	103	77.5	819.7	1118.3	187.0
	Median	8.9	14	216	34.3	685.8	1336.3	594.5

The dataset is split into three sets: a training set from February 1, 2022 to September 8, 2023 (584 days, 60 %), a validation set from September 9, 2023 to January 31, 2024 (145 days, 15 %), and a test set from February 1, 2024 to September 30, 2024 (244 days, 20 %). All features and labels are min-max normalized to a range between 0 and 1.

3. Results

First, the importance of individual features for the different datasets is analysed (Section 3.1). This will determine in which datasets occupancy has a high influence and how large its influence is compared to the other features. Section 3.2 then analyses the load forecast with and without occupancy for the different hotel datasets.

3.1. Feature Importance for the different datasets

FI was calculated using a random forest model and the results are shown in Table 2. The labels for the model were different heating $\dot{Q}$ and the electric loads P_{el}. Of the weather features, T_{out} is the most important for forecasting heating load, while GHI is the dominant feature for electrical power. Among the time features, the week of the year plays an important role for heating forecast, while the hour of the day is most relevant for electric power. The importance

Journal of Physics: Conference Series **3140** (2025) 052023 doi:10.1088/1742-6596/3140/5/052023

of occupancy varies across datasets, ranging from 3.0 % to 14.7 %, with notably lower values for heating load in the guest room wing ($\dot{Q}_{\mathrm{GRW}}$) and total hotel load in Hotel 2.

Table 2. Relative permutation feature importance in percent for the different datasets

Feature	Hotel 1				Hotel 2			
	$\dot{Q}_{\mathrm{SP}}$	$\dot{Q}_{\mathrm{GRW}}$	$\dot{Q}_{\mathrm{Hotel}}$	P_{el}	$\dot{Q}_{\mathrm{SP}}$	$\dot{Q}_{\mathrm{GRW}}$	$\dot{Q}_{\mathrm{Hotel}}$	P_{el}
T_{out}	50.7	59.6	57.3	7.5	44.5	64.2	68.4	6.7
GHI	6.6	3.6	10.4	47.0	8.5	8.5	8.5	41.6
Hour	7.8	2.6	10.6	29.5	9.1	19.1	14.2	33.3
Day of week	2.7	3.7	2.3	1.7	3.0	1.1	1.2	1.5
Week of year	20.0	15.7	10.6	5.3	22.9	3.9	4.2	6.3
Occupancy	12.1	14.7	8.8	9.0	12.0	3.0	3.6	10.7

3.2. Impacts of occupancy data on load forecast

The LSTM and the LSTM-based ED model are applied to forecast heating and electricity loads during the test period for eight datasets from the two hotels. A maximum of 50 epochs were conducted for both the LSTM and the LSTM-based ED model, after which the model with the lowest validation MSE was selected. The optimal hyperparameters were identified through a grid search. The learning rate was set to 0.0001, and the Adam optimiser was used. A batch size of 256 was applied.

Table 3 shows the NRMSE over the 24-hour forecast horizon for the LSTM model, while Table 4 shows the results for the ED model. In both tables, forecast accuracy is evaluated with and without the inclusion of occupancy as a feature. The relative change in NRMSE between these two cases is also reported to quantify the influence of occupancy information on forecast accuracy. For most of the heating load datasets in the LSTM model (see Table 3), the inclusion of occupancy has only a marginal effect, resulting in slight improvements in a few cases. The model appears to be struggling to identify a significant correlation between electricity load and occupancy, as evidenced by the consistent rise in NRMSE when utilising this data. This means that consumption is largely independent of the number of guests and is instead determined by periodic base loads, which are better captured by time characteristics. Consequently, the occupancy feature introduces more statistical noise than valuable information.

Table 3. NRMSE for the various hotels and usage types without and with the feature occupancy for the LSTM model

NRMSE/%	Hotel 1				Hotel 2			
	$\dot{Q}_{\mathrm{SP}}$	$\dot{Q}_{\mathrm{GRW}}$	$\dot{Q}_{\mathrm{Hotel}}$	P_{el}	$\dot{Q}_{\mathrm{SP}}$	$\dot{Q}_{\mathrm{GRW}}$	$\dot{Q}_{\mathrm{Hotel}}$	P_{el}
Without Occupancy	4.629	8.028	6.311	19.266	4.444	4.326	4.364	14.926
With Occupancy	4.640	8.041	6.228	19.879	4.461	4.322	4.346	15.309
Relative Difference	0.238	0.162	-1.314	3.182	0.382	-0.092	-0.412	2.565

For the ED model, the average NRMSE across all heating datasets is 4.91 % without occupancy and 4.90 % with it. For electricity forecasting, the average NRMSE is 7.20 % without occupancy and 7.07 % with occupancy. Using occupancy data slightly improves the accuracy of both heating and electricity load forecasts, with a more noticeable benefit in electricity prediction. The mean values reported are based on an average across both configurations with and without time features.

The influence of occupancy is greater when time features are excluded, leading to stronger improvements (e.g., a 3.8 % NRMSE reduction for $\dot{Q}_{\mathrm{SP}}$ in Hotel 1) or degradations, depending on the label. This suggests that time features already encode many occupancy-related patterns. When included, occupancy improves forecasts consistently only for $\dot{Q}_{\mathrm{SP}}$, as shown in Table 4. The heating forecast showed that occupancy had a greater impact on Hotel 1, which had a higher mean occupancy (11.9 %) versus Hotel 2 (6.2 %). For electricity (P_{el}), Hotel 2 shows a slightly larger improvement in forecast accuracy, which aligns with its higher occupancy feature importance (10.7 % vs. 9.0 %).

Table 4. NRMSE for the various hotels and usage types without and with the feature occupancy for the LSTM-based ED model, grouped by presence of time features.

NRMSE %	with time feature	Hotel 1 $\dot{Q}_{\mathrm{SP}}$	Hotel 1 $\dot{Q}_{\mathrm{GRW}}$	Hotel 1 $\dot{Q}_{\mathrm{Hotel}}$	Hotel 1 P_{el}	Hotel 2 $\dot{Q}_{\mathrm{SP}}$	Hotel 2 $\dot{Q}_{\mathrm{GRW}}$	Hotel 2 $\dot{Q}_{\mathrm{Hotel}}$	Hotel 2 P_{el}
Without Occupancy	yes	4.029	7.558	4.965	7.770	4.372	4.366	4.400	6.795
With Occupancy	yes	4.027	7.606	5.043	7.777	4.363	4.368	4.322	6.945
Relative Difference	yes	-0.050	0.635	1.570	0.090	-0.206	0.046	-1.773	2.208
Without Occupancy	no	4.084	7.365	4.916	7.581	4.339	4.292	4.278	6.672
With Occupancy	no	3.928	7.330	4.826	7.261	4.343	4.334	4.329	6.282
Relative Difference	no	-3.820	-0.475	-1.830	-4.221	0.092	0.979	1.192	-5.845

However, for $\dot{Q}_{\mathrm{GRW}}$ and P_{el}, the inclusion of occupancy sometimes leads to a slight deterioration, depending on the hotel and model configuration. For example, while occupancy has higher FI for $\dot{Q}_{\mathrm{GRW}}$ in Hotel 1 (14.7 %) than in Hotel 2 (3.0 %), the performance impact is minimal or even slightly positive in Hotel 1 and slightly negative in Hotel 2. For P_{el}, occupancy has higher importance in Hotel 2 (10.7 % vs. 9.0 %) and also yields a slightly larger improvement in forecast accuracy.

The heating and electrical load forecasts for the entire Hotel 1, without time features, are shown in Figure 1.

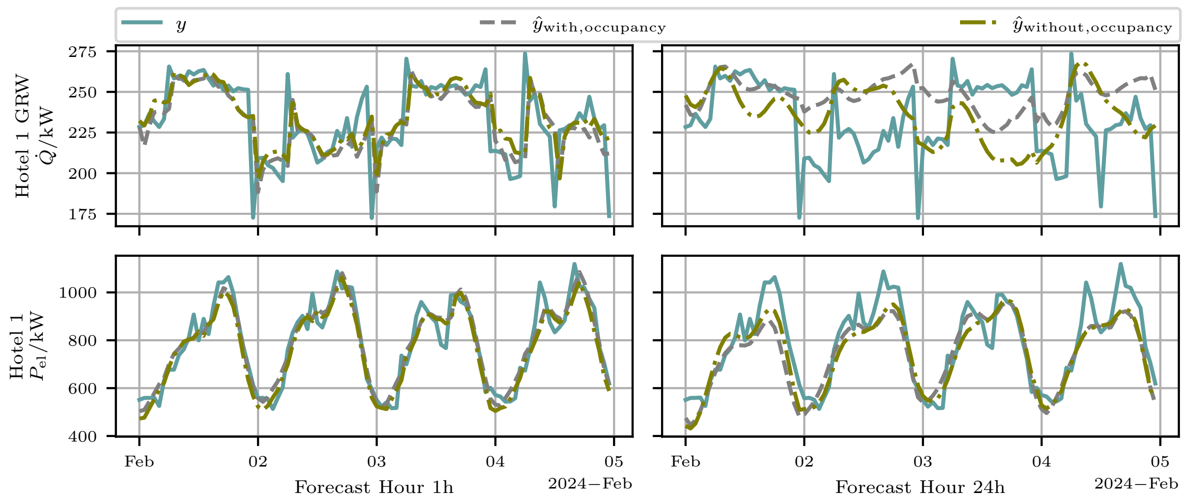

Figure 1. Forecasting heat load for $\dot{Q}_{\mathrm{GRW}}$ and electrical load with and without occupancy (without time features)

The forecasts are presented for both 1-hour and 24-hour horizons, with and without the inclusion of occupancy. In all cases, the 1-hour forecasts are closer to the actual values than the 24-hour forecasts. Both the heating and electricity load show a clear periodic pattern, with the lowest values occurring around midnight.

4. Conclusion

This paper examines if the inclusion of occupancy data improves the prediction accuracy of heating and electricity loads in hotels using LSTM-based models. By applying LSTM-based ED and LSTM architectures to real data from two hotels with different usage patterns, the study shows that the effect of occupancy is context dependent.

For heating forecasts, the inclusion of occupancy data reduced the average NRMSE from 4.91 % to 4.90 %, while for electricity it decreased from 7.20 % to 7.07 %. The impact of occupancy was stronger when time features were excluded and was generally more beneficial in cases where occupancy showed high FI. These results suggest that occupancy should not be included by default, but evaluated based on the prediction target and feature relevance.

Future work should apply this approach to other types of building use, such as office buildings, where occupancy could be more significant. The results also indicate that hotel load profiles frequently exhibit a substantial fixed base load, reducing the impact of user-dependent factors.

Acknowledgements

This work is funded by the German Federal Ministry for Economic Affairs and Climate Action in the framework of the research program EnOB:ML-EBESR 03EN1076B. The support is gratefully acknowledged. We also thank empact GmbH for providing the heating and electricity demand data used in this study. The data were collected by empact GmbH as part of the research project Pilotprogramm Einsparzähler, which is likewise funded by the Federal Ministry for Economic Affairs and Climate Action.

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